

1. $\int \ln(x) \arcsen(x) dx$

por partes:

$$\begin{aligned} \ln x = u \quad \arcsen x = dv \\ \frac{dx}{x} = du \quad x \arcsen x + \sqrt{1-x^2} = v \end{aligned} \Rightarrow I = x \ln x \arcsen x + \ln x \sqrt{1-x^2} - \int \arcsen x dx - \int \frac{\sqrt{1-x^2}}{x} dx$$

$$= \ln(x^2) \arcsen x + \sqrt{1-x^2} \ln(x) - x \arcsen x - \sqrt{1-x^2} - \ln \left| \frac{1}{\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{x} - \sqrt{1-x^2} \right| + C$$

$$\int \arcsen x dx = \int \theta \cos \theta d\theta = \theta \sin \theta + \cos \theta = x \arcsen x + \sqrt{1-x^2}$$

$$\begin{aligned} x &= \sin \theta \\ dx &= \cos \theta d\theta \end{aligned}$$

$$\int \frac{\sqrt{1-x^2}}{x} dx = \int \frac{\cos \theta \cos \theta d\theta}{\sin \theta} = \int (\cos \theta - \sin \theta) d\theta = \ln |\cos \theta - \sin \theta| + \cos \theta = \ln \left| \frac{1}{\sqrt{1-x^2}} - \frac{\sqrt{1-x^2}}{x} \right| + \sqrt{1-x^2}$$

$$\begin{aligned} x &= \sin \theta \\ dx &= \cos \theta d\theta \end{aligned}$$

2. $\int e^x \left(\frac{1-5x+x^2+\sen(x)-5x\cos(x)+x^2\cos(x)}{1+\cos(x)} \right) dx$

$$\int e^x \left(\frac{-5x(1+\cos x) + x^2(1+\cos x) + (1+\sen x)}{1+\cos x} \right) dx = \int (x^2-5x)e^x dx + \int \frac{(1+\sen x)e^x}{1+\cos x} dx$$

$$= (x^2-5x)e^x - (2x-5)e^x + 2e^x + e^x \tan \frac{x}{2} + C$$

$$\int \frac{(1+\sen x)e^x}{1+\cos x} dx = \int \frac{e^x}{1+\cos x} dx + \int \frac{e^x \sen x}{1+\cos x} dx = e^x \tan \frac{x}{2}$$

$$I_1 = \int \frac{e^x}{2 \cos^2 \frac{x}{2}} dx = \frac{1}{2} \int e^x \sec^2 \frac{x}{2} dx = e^x \tan \frac{x}{2} - \int e^x \tan \frac{x}{2} dx$$

$$I_2 = \int \frac{e^x \sen x}{2 \cos^2 \frac{x}{2}} dx = \int e^x \tan \frac{x}{2} dx$$

3. $\int \frac{(10-\cos^2(x)-4\sen(x))\cos(x)}{(\sen^2(x)+1)(\sen^2(x)-2\sen(x)+5)} dx$

$$I = \int \frac{(\sen^2 x - 4 \sen x + 9) \cos x}{(\sen^2 x + 1)(\sen^2 x - 2 \sen x + 5)} dx = \int \frac{(u^2 - 4u + 9)}{(u^2 + 1)(u^2 - 2u + 5)} du = 2 \arctan u - \frac{1}{2} \arctan \left(\frac{u-1}{2} \right) + C$$

$$\sen x = u \Rightarrow 2 \arctan(\sen x) - \frac{1}{2} \arctan \left(\frac{\sen x - 1}{2} \right) + C$$

* por fracciones parciales:

$$\frac{u^2 - 4u + 9}{(u^2 + 1)(u^2 - 2u + 5)} = \frac{Au + B}{u^2 + 1} + \frac{Cu + D}{u^2 - 2u + 5} = \frac{Au^2 - 2Au + 5Au + Bu^2 - 2Bu + 5B + Cu^2 + Cu + D}{(u^2 + 1)(u^2 - 2u + 5)}$$

$$u^2 - 4u + 9 = u^3(A+C) + u^2(-2A+B+D) + u(5A-2B+C) + 5B+D$$

$$\begin{cases} A+C=0 \\ -2A+B+D=1 \\ 5A-2B+C=-4 \\ 5B+D=9 \end{cases} \Rightarrow \begin{cases} A=0 \\ B=2 \\ C=0 \\ D=-1 \end{cases}$$

4. $\int \frac{dx}{x(\ln(x)-1)^3(\ln(x)+2)^4}$

Integral de Uris: $\int \frac{dx}{(x+a)^m(x+b)^n} = \frac{1}{(b-a)^{m+n-1}} \int \frac{(1-t)^{m+n-2}}{t^m} dt$

haciendo $t = \frac{x+a}{x+b}$

$$I = \int \frac{dx}{(\ln x - 1)^3 (\ln x + 2)^4} = \frac{1}{3^6} \int \frac{(1-t)^5}{t^3} dt = \frac{1}{3^6} \int \left(\frac{1}{t^3} - \frac{5}{t^2} + \frac{10}{t} - 10t + t^2 \right) dt$$

$$\text{haciendo } t = \frac{\ln x - 1}{\ln x + 2} \quad (1-t)^5 = 1 - 5t + 10t^2 - 10t^3 + 5t^4 - t^5$$

$$I = \frac{1}{3^6} \left[-\frac{(\ln x - 1)^2}{2} + 5 \frac{(\ln x - 1)}{\ln x + 2} + 10 \ln \left| \frac{\ln x - 1}{\ln x + 2} \right| - 10 \frac{(\ln x - 1)}{\ln x + 2} + \frac{5}{2} \frac{(\ln x - 1)^2}{\ln x + 2} - \frac{1}{3} \frac{(\ln x - 1)^3}{\ln x + 2} \right] + C$$

5. $\int \frac{e^{4x} + 3e^{3x} - e^{2x} + 2e^x}{(e^{2x} + 4e^x + 3)^3} dx$

$$e^x = u \Rightarrow I = \int \frac{u^3 + 3u^2 - u + 2}{(u^2 + 4u + 3)^3} du = \int \frac{u^3 + 4u^2 + 3u}{(u^2 + 4u + 3)^3} du - \int \frac{u^2 + 4u + 3}{(u^2 + 4u + 3)^3} du + 5 \int \frac{du}{(u^2 + 4u + 3)^3}$$

$$* I_1 = \int \frac{u + 2 - 2}{(u^2 + 4u + 3)^2} du = \frac{1}{2} \int \frac{(2u + 4)}{(u^2 + 4u + 3)^2} du - 2 \int \frac{du}{(u^2 + 4u + 3)^2} = -\frac{1}{2} (u^2 + 4u + 3)^{-1} - 2 I_2$$

$$* I_2 = \int \frac{du}{(u+3)^2(u+1)^2} = -\frac{1}{8} \left(\frac{u+1}{u+3} \right)^{-1} \frac{1}{4} \ln \left| \frac{u+1}{u+3} \right| + \frac{1}{8} \left(\frac{u+1}{u+3} \right)$$

$$* I_3 = \int \frac{du}{(u+1)^3(u+3)^3} = \frac{1}{32} \left(-\frac{1}{2} \left(\frac{u+1}{u+3} \right)^2 + 4 \left(\frac{u+1}{u+3} \right)^{-1} + 6 \ln \left| \frac{u+1}{u+3} \right| - 4 \left(\frac{u+1}{u+3} \right) + \frac{1}{2} \left(\frac{u+1}{u+3} \right)^3 \right)$$

6. $\int \frac{\cos(x)}{1-3\sen(x)+2\cos(x)} dx$

$$\begin{aligned} x &= 2t \\ dx &= 2dt \end{aligned} \quad I = \frac{1}{2} \int \frac{\cos 2t}{1-3\sen 2t+2\cos 2t} dt$$

$$2I = \int \frac{\frac{1+\cos 2t}{2}}{1-\frac{6\sen t}{2}+\frac{2+\cos 2t}{2}} dt = \int \frac{1+\cos 2t}{3-6\sen t+2+\cos 2t} dt = \int \frac{1+\cos 2t-1}{1+\cos 2t-3\sen t} dt$$

$$\begin{aligned} \tan t &= u \\ dt &= \frac{du}{u^2+1} \end{aligned}$$

$$2I = \int \frac{u^2-1}{(u^2+1)(u^2+6u-3)} dt = \frac{1}{18} \int \frac{3u+2}{u^2+1} du - \frac{1}{18} \int \frac{3u+1}{u^2+6u-3} du = \frac{3}{26} \ln |u| + \frac{2}{13} \arctan u - \frac{3}{26} \ln |u^2+6u-3| + \frac{1}{52} \ln \left| \frac{u+3}{u+2} \right| + C$$

fracciones parciales:

$$\frac{u^2-1}{(u^2+1)(u^2+6u-3)} = \frac{Au+B}{u^2+1} + \frac{Cu+D}{u^2+6u-3}$$

$$A = \frac{3}{18}, B = \frac{2}{18}, C = -\frac{3}{18}, D = -\frac{2}{18}$$

$$I = \frac{3}{52} \ln \left| \tan \left(\frac{x}{2} \right) + 1 \right| + \frac{2}{26} - \frac{3}{52} \ln \left| \tan \left(\frac{x}{2} \right) + 6 \tan \left(\frac{x}{2} \right) - 3 \right| + \frac{1}{5040} \ln \left| \frac{\tan \left(\frac{x}{2} \right) + 3}{\tan \left(\frac{x}{2} \right) + 2} \right| + C$$